LLM Transmission 8GB — Comprehensive Findings Report

Generated: September 02, 2026

Written in the Phosphorus remedy personality — The Charismatic Communicator

NVIDIA Jetson Nano 8GB | llama.cpp CUDA | 27 models benchmarked | 2,916 scored rows

Welcome! This report shares everything we learned from systematically testing 27 local LLMs on an 8GB Jetson edge device. We swept temperature across 7 values, top_k across 3 values, discovered and fixed two major chat template bugs, and built a routing service that uses these findings to pick the best model for any prompt — all locally, zero cloud tokens.

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1. The Story — How We Benchmarked 27 Models

The Setup

Picture this: an 8GB NVIDIA Jetson sitting on a desk, its GPU fully offloaded with llama.cpp CUDA, flash attention cranked on, the GUI disabled to squeeze every megabyte of RAM for model inference. That's the stage. On it, we loaded 27 quantized language models — from tiny 1B parameters up to a 7B DeepSeek R1 — and ran each through a carefully designed gauntlet of 12 prompts spanning 6 categories.

The 12-Prompt Gauntlet

How We Scored

Every prompt was scored on a 0-10 scale. For code, we checked correctness, edge cases, and clean style. For poetry, we verified strict iambic pentameter meter — no easy feat for a 3B model! For function calls, the make-or-break criterion was whether the model produced valid JSON tool-call format, not just prose that described what it would do. That distinction matters: most models can describe a tool call in English, but far fewer can actually produce the structured JSON that a real agent system needs.

The Sweep Strategy

We followed one golden rule: quality before speed. A model that generates wrong answers quickly is useless. So we optimized one variable at a time:

Constants We Held Steady

top_p = 0.9 | repeat_penalty = 1.1 | max_tokens = 2048 | GPU layers = 999 (all) | Flash attention = on. These stayed fixed so we could isolate the effect of each variable we swept.

2. The Leaderboard — All 27 Models Ranked

Here it is — the full picture. Every model, ranked by its overall average across all 12 prompts at its best temperature with k=40. The group columns show where each model shines or stumbles.

3. Temperature Sweep — Finding Each Model's Sweet Spot

We tested every model at 7 temperatures, from frozen deterministic (0.0) to wild creative (1.0). The result? There is no universal best temperature. The distribution tells the story:

The pattern is beautiful: reasoning models (DeepSeek R1, Ministral Reasoning) crave low temperatures (0.0-0.3) where deterministic output helps them think step by step. Creative models (Granite 4.1, Qwen 3 1.7B) blossom at high temperatures (1.0) where randomness fuels imaginative connections. This is why our transmission service stores a per-model locked temperature — a global default would shortchange half the models.

4. Top-K Sweep — Confirming the Universal Default

After locking each model's best temperature, we swept top_k across 3 values. The verdict was decisive:

k=40 is the champion — 78% of models prefer it. The 4 models that prefer k=20 lose less than 0.3 points at k=40, so we use k=40 as the universal default. One setting to rule them all, and in the routing table bind them.

5. The Great Template Discovery — DeepSeek & E2B Fixes

This is the story that made the biggest difference. Two model families were using the wrong chat template — and the fix transformed garbage into gold.

DeepSeek R1 Distills: deepseek to chatml

DeepSeek R1-Distill models are Qwen-architecture models fine-tuned by DeepSeek. The name says 'DeepSeek,' so we used the 'deepseek' chat template. Wrong! These models use ChatML format. With the wrong template, they hallucinated about 'DeepSeek company' instead of answering prompts. The fix was simple but dramatic:

E2B MatFormer Models: gemma to chatml

Gemma 3n E2B and Gemma 4 E2B are MatFormer models. The name says 'Gemma,' so we used the 'gemma' chat template. Again wrong! With the gemma template, gemma3n-e2b hallucinated about 'Gemini 1.5 Pro vs GPT-4 Turbo' and gemma4-e2b output only '_4'. Both were completely broken. Switching to chatml with --jinja brought them to life:

The lesson shines bright: a model's name does not determine its chat template. DeepSeek R1 distills use chatml. E2B MatFormer models use chatml. Always verify the template matches the architecture, not the brand. Both fixes are now protected by test assertions in the transmission service to prevent regression.

6. Per-Model Findings — 27 Models, Each with Ideal Settings

Here's where we get up close and personal with each model. For every one of the 27, you'll find its ideal settings, per-prompt score breakdown, and what we observed during testing.

#1 — hermes3-3b-q5

Overall: 7.8/10 | Coding/Math: 9.8 | Creative/Poetry: 9.0 | Function Calls: 6.2 | Speed: 19.4 tok/s

- Excellent at coding & math

- Excellent at creative writing & poetry

- Recommended for production use

#2 — smallthinker-3b

Overall: 7.6/10 | Coding/Math: 8.5 | Creative/Poetry: 5.5 | Function Calls: 7.7 | Speed: 20.1 tok/s

- Strong function-call format compliance

- Reasoning model — produces thinking blocks, needs --jinja flag

- Recommended for production use

#3 — deepseek-r1-7b

Overall: 7.5/10 | Coding/Math: 10.0 | Creative/Poetry: 7.0 | Function Calls: 6.3 | Speed: 9.7 tok/s

- Excellent at coding & math
- Slow generation (9.7 tok/s) — may timeout on long prompts
- Reasoning model — produces thinking blocks, needs --jinja flag
- Recommended for production use
- DeepSeek R1 distill — chatml template fix applied (was deepseek, +43%/+158%)

#4 — hermes3-3b-q4

Overall: 7.5/10 | Coding/Math: 9.8 | Creative/Poetry: 6.5 | Function Calls: 6.3 | Speed: 19.7 tok/s

- Excellent at coding & math
- Recommended for production use

#5 — qwen2.5-3b

Overall: 7.5/10 | Coding/Math: 9.0 | Creative/Poetry: 8.5 | Function Calls: 6.2 | Speed: 20.1 tok/s

- Excellent at coding & math

- Recommended for production use

#6 — qwen3.5-2b

Overall: 7.5/10 | Coding/Math: 9.8 | Creative/Poetry: 9.0 | Function Calls: 5.5 | Speed: 24.9 tok/s

- Excellent at coding & math

- Excellent at creative writing & poetry

- Reasoning model — produces thinking blocks, needs --jinja flag

- Recommended for production use

#7 — granite4.1-3b

Overall: 7.4/10 | Coding/Math: 9.2 | Creative/Poetry: 8.5 | Function Calls: 5.8 | Speed: 18.9 tok/s

- Excellent at coding & math

- Recommended for production use

#8 — ministral-3b-reasoning

Overall: 7.2/10 | Coding/Math: 9.5 | Creative/Poetry: 9.5 | Function Calls: 5.0 | Speed: 18.7 tok/s

- Excellent at coding & math

- Excellent at creative writing & poetry

- Reasoning model — produces thinking blocks, needs --jinja flag

- Recommended for production use

#9 — qwen3-1.7b

Overall: 7.2/10 | Coding/Math: 9.8 | Creative/Poetry: 5.5 | Function Calls: 6.0 | Speed: 30.8 tok/s

- Excellent at coding & math

- Fast generation (30.8 tok/s)

- Recommended for production use

#10 — smollm3

Overall: 7.2/10 | Coding/Math: 9.2 | Creative/Poetry: 7.0 | Function Calls: 5.8 | Speed: 20.8 tok/s

- Excellent at coding & math

- Recommended for production use

#11 — gemma3n-e2b

Overall: 7.1/10 | Coding/Math: 9.2 | Creative/Poetry: 9.5 | Function Calls: 4.8 | Speed: 21.5 tok/s

- Excellent at coding & math
- Excellent at creative writing & poetry
- Reasoning model — produces thinking blocks, needs --jinja flag
- Recommended for production use
- MatFormer (E2B) model — chatml template fix applied (was gemma, +273%/+650%)

#12 — granite3.2-2b

Overall: 7.0/10 | Coding/Math: 9.5 | Creative/Poetry: 7.0 | Function Calls: 5.3 | Speed: 26.6 tok/s

- Excellent at coding & math
- Fast generation (26.6 tok/s)
- Recommended for production use

#13 — granite4.2-3b

Overall: 6.9/10 | Coding/Math: 9.5 | Creative/Poetry: 6.5 | Function Calls: 5.3 | Speed: 18.9 tok/s

- Excellent at coding & math

#14 — qwen2.5-coder-3b

Overall: 6.9/10 | Coding/Math: 9.5 | Creative/Poetry: 8.5 | Function Calls: 4.7 | Speed: 20.0 tok/s

- Excellent at coding & math

#15 — granite4-3b

Overall: 6.8/10 | Coding/Math: 9.5 | Creative/Poetry: 6.5 | Function Calls: 5.0 | Speed: 18.9 tok/s

- Excellent at coding & math

#16 — stablelm-zephyr

Overall: 6.8/10 | Coding/Math: 9.8 | Creative/Poetry: 8.0 | Function Calls: 4.3 | Speed: 26.4 tok/s

- Excellent at coding & math

- Fast generation (26.4 tok/s)

#17 — lfm2.5-2.6b

Overall: 6.6/10 | Coding/Math: 10.0 | Creative/Poetry: 5.5 | Function Calls: 4.7 | Speed: 23.8 tok/s

- Excellent at coding & math

#18 — gemma4-e2b

Overall: 6.0/10 | Coding/Math: 9.2 | Creative/Poetry: 9.5 | Function Calls: 2.7 | Speed: 23.4 tok/s

- Excellent at coding & math

- Excellent at creative writing & poetry

- Struggles with function-call JSON — produces prose instead

- Reasoning model — produces thinking blocks, needs --jinja flag

- MatFormer (E2B) model — chatml template fix applied (was gemma, +273%/+650%)

#19 — deepseek-r1-1.5b

Overall: 5.3/10 | Coding/Math: 9.2 | Creative/Poetry: 4.5 | Function Calls: 3.0 | Speed: 32.7 tok/s

- Excellent at coding & math
- Struggles with function-call JSON — produces prose instead
- Fast generation (32.7 tok/s)
- Reasoning model — produces thinking blocks, needs --jinja flag
- DeepSeek R1 distill — chatml template fix applied (was deepseek, +43%/+158%)

#20 — ministral-3b

Overall: 4.9/10 | Coding/Math: 7.2 | Creative/Poetry: 5.5 | Function Calls: 3.2 | Speed: 24.7 tok/s

- Struggles with function-call JSON — produces prose instead

#21 — gemma2-2b

Overall: 4.5/10 | Coding/Math: 2.8 | Creative/Poetry: 4.0 | Function Calls: 5.8 | Speed: 22.3 tok/s

#22 — llama3.2-3b

Overall: 2.8/10 | Coding/Math: 1.8 | Creative/Poetry: 5.0 | Function Calls: 2.7 | Speed: 19.7 tok/s

- Struggles with function-call JSON — produces prose instead

- Very poor — not recommended for production use

#23 — gemma3-1b

Overall: 2.7/10 | Coding/Math: 2.2 | Creative/Poetry: 3.5 | Function Calls: 2.7 | Speed: 31.5 tok/s

- Struggles with function-call JSON — produces prose instead

- Fast generation (31.5 tok/s)

- Very poor — not recommended for production use

#24 — phi3-3.8b

Overall: 2.5/10 | Coding/Math: 2.2 | Creative/Poetry: 4.0 | Function Calls: 2.2 | Speed: 17.7 tok/s

- Struggles with function-call JSON — produces prose instead

- Very poor — not recommended for production use

#25 — llama3.2-1b

Overall: 2.4/10 | Coding/Math: 1.2 | Creative/Poetry: 4.5 | Function Calls: 2.5 | Speed: 42.2 tok/s

- Struggles with function-call JSON — produces prose instead

- Fast generation (42.2 tok/s)

- Very poor — not recommended for production use

#26 — phi4-mini

Overall: 2.3/10 | Coding/Math: 2.5 | Creative/Poetry: 3.0 | Function Calls: 2.0 | Speed: 16.8 tok/s

- Struggles with function-call JSON — produces prose instead

- Reasoning model — produces thinking blocks, needs --jinja flag

- Very poor — not recommended for production use

#27 — llama3.2-3b-new

Overall: 2.2/10 | Coding/Math: 1.5 | Creative/Poetry: 3.5 | Function Calls: 2.2 | Speed: 19.7 tok/s

- Struggles with function-call JSON — produces prose instead

- Very poor — not recommended for production use

7. Category Analysis — Who Shines Where

Coding & Math

Structured generation — Python functions, HTML pages, math proofs. This is where most models do their best work.

Creative & Poetry

Linguistic artistry — iambic pentameter, short stories, imaginative prose. The hardest category to score high.

Function Calls

Structured tool-use JSON — terminal commands, file operations, SQLite queries, email drafts. The weakest category across the board.

8. Building the Transmission — From Data to Service

All that benchmark data is great, but what do you actually do with 2,916 scored rows? You build a service that puts them to work. That's the LLM Transmission 8GB — a prompt router that classifies what you're asking, shows you the best models for that type of task, and runs your prompt with the selected model's optimal settings.

The Architecture

The pipeline is beautifully simple: Prompt arrives -> Classifier matches keywords in <1ms (no LLM needed!) -> Routing table presents top 6 models for the detected category -> Operator picks one (or accepts #1) -> llama-cli runs with that model's locked-best temp and k=40 -> Output is cleaned and returned with stats. Zero cloud tokens.

Step 1: Collect the Data

27 models, 12 prompts, 7 temperatures, 3 top_k values. Every run scored 0-10. The result: a CSV with 2,916 rows that tells us exactly which model is best at what, and what settings to use.

Step 2: Lock the Best Parameters

For each model, we locked the best temperature (the one that produced the highest average score). Then we confirmed k=40 as the universal top_k. Then we discovered and fixed the two template bugs. Each step was one variable at a time, quality before speed.

Step 3: Classify Without an LLM

The classifier uses pure Python keyword matching and heuristics — it runs in under 1 millisecond with zero model loading. It scores the prompt against keyword indices for 4 categories: coding_math (python, html, prove), creative_poetry (poem, iambic, story), function_calls (terminal, sqlite, email), and general_purpose (what, how, explain). Heuristic boosts catch edge cases like 'write a function' without a language name.

Step 4: Build the Routing Table

For each category, we ranked all 27 models by their group-average quality score and took the top 6. Each entry carries everything the service needs: quality, speed, file size, chat template, thinking flag, best temperature, top_k, and model file path.

Step 5: Present and Execute

When a prompt arrives, the service classifies it, shows the operator a clean table of 6 options with quality, speed, size, thinking flag, and a QxS balance metric, runs the chosen model, cleans the output, and reports how many tokens were generated — and the satisfying fact that zero cloud tokens were spent.

9. Routing Table — Top 6 Models Per Category

Coding Math

Python code, HTML pages, math proofs — structured generation requiring precision

Creative Poetry

Creative writing, poetry, iambic pentameter, stories — requires linguistic artistry

Function Calls

Hermes tool calls: terminal, write_file, read_file, web_search, sqlite, email — structured tool-use format

General Purpose

General-purpose prompts, questions, explanations — fallback for unclassified prompts

10. What We Learned and Where We're Going

The Big Takeaways

- hermes3-3b-q5 is the overall champion (7.8/10) — versatile, strong everywhere, the best general-purpose model.

- lfm2.5-2.6b achieves perfect coding & math (10.0/10) at 23.8 tok/s — the coding specialist.

- gemma3n-e2b rose from 1.9 to 7.1 after the chatml fix — a 273% improvement that shows how crucial templates are.

- gemma4-e2b rose from 0.8 to 6.0 — a 650% improvement from the same one-line fix.

- ministral-3b-reasoning is the best creative writer (9.5/10) — the only model to score perfect 10 on iambic pentameter.

- smallthinker-3b is the best function caller (7.7/10) — the only model that reliably produces valid tool-call JSON.

- DeepSeek R1 7B is now #1 for coding (10.0, tied with LFM) after the chatml template fix — a 158% improvement.

- k=40 is the universal optimal top_k (78% of models) — one setting that works for almost everyone.

- No single temperature works — per-model locked temps are essential. 7 peak at 0.3, 7 at 1.0.

- Function calls remain the hardest category — no model above 7.7/10. Most produce prose, not JSON.

- Models under 2B parameters (gemma3-1b, llama3.2-1b, phi3-3.8b) score below 3.0 — not production-ready.

- A model's name does not determine its chat template. DeepSeek uses chatml. E2B uses chatml. Always verify.

What's Next

- Top-P sweep: Test top_p values (0.8, 0.85, 0.9, 0.95, 1.0) at locked temps and k=40.

- Repeat penalty sweep: Test 1.0, 1.05, 1.1, 1.15, 1.2 for per-model optimization.

- Context window tuning: Find the sweet spot between 8K, 16K, and 32K for 8GB VRAM.

- Function-call improvement: Explore system prompts or fine-tuning to help models produce better JSON.

- New model testing: Add new releases as they arrive (Qwen 3.5, Granite 5, etc.).

- Web interface: A Flask UI for non-CLI usage.

- Auto-routing mode: Skip operator selection, auto-pick #1 based on quality score.

The beauty of this project is that every finding is backed by real data — 2,916 scored runs on actual hardware. No speculation, no hand-waving. Just models, prompts, and the numbers they produced.

And now, all of that knowledge lives inside a routing service that puts it to work every time you submit a prompt.